

Global Energy Storage

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Battery Weekly 5 October

America

[Vianode seeks partner for cooperation or acquisition- electrive.com](#) Vianode, the Norwegian synthetic graphite producer, is seeking a strategic partner for its next expansion phase and is open to options ranging from cooperation to a full acquisition. The company operates a 2,000-tonne/year graphite plant in Norway and is developing a second facility in Ontario, Canada, with GM as its anchor customer and production targeted for 2029. Vianode has engaged JPMorgan to support the search as it seeks additional capital and industrial expertise, with the expansion supporting efforts to build a non-Chinese battery-material supply chain amid China's roughly 95% share of global battery-grade graphite supply.

[Ultium Cells retrofits Tennessee plant for LMR cell production- electrive.com](#) Ultium Cells, the battery JV between General Motors and LG Energy Solution, will upgrade its Spring Hill, Tennessee plant to mass-produce prismatic lithium-manganese-rich (LMR) cells for future GM EVs from 2028. The LMR technology is expected to deliver 33% higher energy density than LFP at a comparable cost, providing a middle ground between lower-cost LFP and higher-performance nickel-rich batteries. The plant will ultimately produce LMR, LFP and high-nickel chemistries across pouch and prismatic formats, with combined investments in its battery programs and facility upgrades reaching US\$1 billion by 2030 and creating 500 additional jobs.

[PowerCo postpones battery cell production in Canada- electrive.com](#) PowerCo, Volkswagen's battery subsidiary, has delayed the start of production at its St. Thomas, Ontario gigafactory from 2027 to 2029, citing evolving market demand, battery technology developments and Volkswagen's long-term strategy. The plant was originally designed for up to 90 GWh of annual capacity and will primarily supply batteries for Volkswagen Group EVs produced in North America. Construction is continuing despite the delay, with PowerCo saying the additional time will allow it to integrate next-generation battery technology and retain flexibility to scale capacity as demand evolves. The facility remains a key part of Volkswagen's North American battery strategy.

[LG Energy Solution to Supply Tesla-Style Batteries to Mercedes-Benz EVs.- thelec.net](#) LG Energy Solution plans to supply new 46100 cylindrical battery cells to Mercedes-Benz for next-generation EVs from 2028, with production expected at its Wrocław, Poland facility. The 46100 format, measuring 46 mm in diameter and 100 mm in height, expands LGES's 46-series lineup, while the broader Wrocław 46-series project targets around 11.5–12 GWh of annual capacity from 2028. LGES is also considering bringing module and pack assembly in-house for cylindrical batteries, which could deepen its role in Mercedes-Benz's battery supply chain beyond cell manufacturing.

[LG Energy Solution, US Startup indigoTech to Collaborate on 46-Series Battery Supply.- thelec.net](#) LG Energy Solution has signed an MoU with US electric commercial-vehicle startup indigoTech to explore supplying 46-series NCM cylindrical batteries from 2027 to 2030 for the startup's upcoming Flow Ride and Flow Cargo vehicles. The companies will also collaborate on battery-vehicle integration to improve driving range and charging performance. The partnership expands LGES's 46-series customer base in North America and targets growing applications across logistics, last-mile delivery and ride-hailing, following a 1.5x YoY increase in LGES cylindrical battery shipments in 2026.

[AI Data Centers Lift US Energy Storage Market, Wood Mackenzie Says.- thelec.net](#) Wood Mackenzie expects annual US energy storage installations to grow more than 50% in MW terms over the next five years, reaching cumulative capacity of 207 GW / 715 GWh by 2031. The US installed a record 5.4 GW / 18.9 GWh of battery storage in 2026, while average system duration increased from 2.8 hours to 3.5 hours YoY, reflecting a shift toward longer-duration systems. Utility-scale storage is forecast to grow by an average 8% annually over the next five

years, supported by rising electricity demand and the need for rapidly deployable capacity, including for expanding data-center loads.

[These Battery Cells Were Tested After 14 Years Of Hard Work. They're Good For Another Decade.- insideevs](#) CATL tested more than 50 LFP cells from China's 63 MWh Zhangbei energy storage project, which operated from 2011 to 2025 without replacing any CATL cells. After nearly 14 years of operation, the cells retained around 85% of their original capacity and could potentially deliver another ~1,000 cycles or 10 years in lower-capacity storage applications. The results demonstrate the strong long-term durability and second-life potential of LFP technology for stationary energy storage.

Asia

[How Geely's New Battery Handles A Megawatt Of Charging Power Without Crazy Degradation.- insideevs](#) Geely Auto Group has introduced its new Ultra Short Blade Battery, capable of peak charging power of around 1,093 kW, with a Lynk & Co 10 demonstrating 10%-97% charging in around 8 minutes 40 seconds. To limit degradation from extreme fast charging, Geely uses shorter cells to reduce ion travel and internal resistance, alongside enhanced electrode structures, lithium pulse technology and AI-based thermal management that predicts battery temperatures 30 seconds in advance and adjusts charging power accordingly. The company has also extended the battery warranty from 8 years/150,000 km to 8 years/200,000 km, highlighting its focus on balancing megawatt-level charging with battery longevity.

[Samsar upcycles EV battery from 1000th Nissan Leaf- electrive.com](#) US battery remanufacturing startup Samsar Resources has repurposed its 1,000th Nissan Leaf battery, returning more than 25 MWh of battery capacity to service through second-life applications. At its Texas facility, Samsar Resources dismantles used EV packs, performs module- and cell-level quality control, and refurbishes suitable components for applications including stationary energy storage, marine and worksite power, with the company claiming its process can capture around 90% of each pack's value. Production has scaled from just 20 packs in 2022 to more than 300 in 2025, with the company targeting a cumulative 2,000 remanufactured packs in 2027.

[China to bring solid-state batteries to mass production by 2030- electrive.com](#) China has released its 2026–2030 battery industry development plan, targeting the initial large-scale application of all-solid-state batteries by 2030, alongside further commercialization of sodium-ion and ultra-fast-charging batteries. The plan also targets 15,000-cycle long-life lithium batteries and parts-per-billion defect rates at leading manufacturers, while promoting battery recycling, lifecycle traceability and industry consolidation. The strategy keeps lithium-ion at the core while developing sodium-ion and flow batteries in parallel, reflecting China's push toward next-generation chemistries, higher manufacturing quality and a more consolidated battery supply chain.

[Gotion targets 400 Wh/kg solid-state batteries- electrive.com](#) Gotion High-Tech has set new commercialization targets for its Jinshi sulfide-based solid-state battery, aiming for 400 Wh/kg energy density and a long-term cost of around US\$150/kWh, improving on its previous 350 Wh/kg benchmark. The company has progressed from a 0.2 GWh pilot line toward a 2 GWh solid-state battery production line, whose design was substantially completed in March 2026. Gotion High-Tech also reported that its solid-state cells completed puncture and high-temperature safety tests without smoke or fire, as the company continues preparations for commercialization.

[Sungrow Promotes Solar and Storage Solutions for Small Commercial Shops.- solarbe.com](#) Sungrow is targeting small commercial shops with rooftop solar solutions, arguing that higher commercial electricity tariffs can make returns attractive despite smaller system sizes. Its SG36/40/50CX inverters offer 36–50 kW capacity, four MPPTs and up to 98.6% efficiency, with global MPPT scanning designed to improve generation under complex rooftop conditions. For businesses operating into the evening, the company also highlights solar-plus-storage as an option to shift daytime generation to nighttime use, positioning integrated solar + storage + EV charging as a potential longer-term solution for commercial premises.

[Global EV Giant BYD Deepens Shaanxi Push as Blade Battery Base Hits Full Production.- yicaglobal.com](#) BYD has started full production at its 16 GWh Xixian battery plant in Shaanxi, one of the company's first manufacturing bases for its second-generation Blade Battery and flash-charging technology. The RMB 7 billion facility can supply batteries for around 700,000 NEVs annually, primarily supporting BYD's local vehicle production, and is expected to generate RMB 8 billion in annual output value. The second-generation Blade Battery supports 10%-70% charging in five minutes, while the plant covers the full manufacturing process from electrode preparation and cell production through pack integration and testing.

Europe

[T&E analysis: sufficient European battery cells by 2030- electrive.com](#) Transport & Environment (T&E) expects European battery cell production to be sufficient by 2030 to meet demand from EVs covered by the EU's proposed Industrial Accelerator Act (IAA) local-content rules, provided planned projects, including medium-confidence projects, materialize. This contrasts with automotive industry concerns that European battery capacity will be insufficient. However, T&E identifies cathode active materials (CAM) and precursor materials (pCAM) as the

more significant supply-chain bottlenecks, highlighting the need for further investment beyond cell manufacturing.

[Battery shortage: Stellantis pauses EV production in France- electrive.com](#) Stellantis will temporarily halt production at its Sochaux and Rennes plants in France in late October due to insufficient supplies of long-range EV batteries from Automotive Cells Company (ACC), its battery JV with Mercedes-Benz and TotalEnergies. Demand for long-range EVs, including the Peugeot E-3008, E-5008 and Citroën ë-C5 Aircross, is exceeding current battery supply as ACC continues to ramp production at its Douvrin factory. The bottleneck highlights ongoing challenges in scaling European battery production despite strong underlying demand for higher-range EV models.

[Industrial Accelerator Act: industry may fail to meet battery demand in EU.- electrive.com](#) A study commissioned by ACEA warns that Europe's battery supply may be insufficient to meet proposed Industrial Accelerator Act (IAA) local-content requirements, even as European cell production is expected to reach 306 GWh by 2032. Across the scenarios assessed, supply remains below demand, potentially leaving around 3 million EVs unable to qualify for "Made in EU" incentives if the IAA takes effect in 2028; for medium- and heavy-duty vehicles, the supply gap could reach 23 GWh by 2032. Key constraints include Europe's limited LFP capacity, higher cell production costs and shortages of cathode and anode active materials, highlighting the need for additional upstream investment and a more gradual localization framework.

[Volkswagen strengthens ties with Gotion with three joint ventures.- electrive.com](#) Volkswagen, PowerCo and Gotion High-Tech plan to establish three battery joint ventures in Spain, Slovakia and Morocco, with total investment of around €3.22 billion, significantly expanding their cooperation in Europe's LFP supply chain. The Spanish Valencia plant will target 29.1 GWh of annual cell capacity, while the Slovakia facility will add 8.4 GWh, primarily supplying Volkswagen's European operations; the Morocco project will produce 100,000 tonnes of LFP cathode materials annually. PowerCo will hold 51% of the Spanish JV, while Gotion High-Tech will hold 51% of the Slovakia and Morocco ventures. The partnership is intended to build a more localized European LFP battery value chain as Volkswagen expects LFP's European market share to rise from around 10% currently to 40–60% by 2030.

[Britain aims to extract lithium from geothermal brine in Cornwall from 2029.- electrive.com](#) Geothermal Engineering Limited (GEL) plans to invest around £43 million in a lithium extraction facility in Cornwall, with commercial production targeted for 2029 and nearly £10 million of support offered through the UK government's DRIVE35 fund. The project will extract around 1,500 tonnes of lithium carbonate annually from geothermal brine, equivalent to the lithium requirement for more than 180,000 typical EV batteries. The project is intended to strengthen the UK's domestic supply of critical battery materials and reduce its reliance on imported lithium as EV and battery production expands.

[Mercedes tests new Gen4 battery cells from ProLogium.- electrive.com](#) Mercedes-Benz is expanding its collaboration with ProLogium, securing priority access to the Taiwanese battery maker's Gen4 solid-state battery cells for comprehensive electrical, thermal and safety testing. The technology uses a non-flammable inorganic electrolyte and ceramic separator and targets higher energy density, fast charging and stronger low-temperature performance, although no series-production decision has been made. The companies have worked together since 2016, with Mercedes also investing in ProLogium in 2022. The partnership could eventually support a localized European supply chain, as ProLogium is building a battery plant in Dunkirk, France, initially targeting 4 GWh of annual capacity and up to 44 GWh longer term.

[LG Chem, Volkswagen to Jointly Develop Next-Generation Automotive Materials.- thelec.net](#) LG Chem and Volkswagen Group have signed an MOU to jointly develop lightweight, high-performance automotive materials using ABS and engineering plastics, targeting lower vehicle weight and improved performance. The companies will also explore greater use of post-consumer recycled (PCR) materials across interior and exterior components of Volkswagen Group vehicles. LG Chem currently has 2.35 million tonnes of annual ABS production capacity globally and is expanding its portfolio of recycled and high-value materials for automotive applications.

[European Battery Makers Turn to Korea's JR Energy Solution for Production Support.- thelec.net](#) European battery producers are looking to South Korea's JR Energy Solution (JRES) for manufacturing support as companies including ACC and Verkor are reportedly operating at below 90% production yields. JRES plans to expand its European foundry business by utilizing idle battery production capacity in France, enabling local production without building a new factory, while contributing its mass-production expertise. Its existing 500 MWh electrode line in Korea is fully utilized, with domestic capacity planned to expand to 2 GWh, while an existing partnership with Morrow Batteries provides additional European manufacturing infrastructure in Norway.

EXHIBIT 1: **Key commodities price performance**

	Price 2-Oct	Ccy	Mkt cap \$bn	P/E 26E	EV /EBITDA 26E	%1D	%1W	%1M	%1Y	1Y px chart
Metal prices (US\$/tonne)										
LiCO (spot)	19,588					0.0%	0%	-16%	91%	
LiOH (spot)	18,098					0.0%	0%	-18%	74%	
LiCO (contract)	18,750					0.0%	-5%	-6%	97%	
LiOH (contract)	18,500					0.0%	1%	-5%	104%	
Cobalt (spot)	38,690					0.0%	-1%	-31%	12%	
Nickel (spot)	15,634					-1.8%	-4%	-6%	2%	
Component prices										
Cathode NMC811						0.0%	-3%	-11%	19%	
Cathode LFP						0.0%	-4%	-17%	40%	
Precursor NMC811						0.0%	-1%	-5%	4%	
Precursor LFP						0.0%	1%	3%	53%	
Artificial Graphite						0.0%	0%	0%	6%	
Natural Graphite						0.0%	0%	0%	0%	
Electrolyte NMC						0.0%	2%	2%	40%	
Electrolyte LFP						0.0%	2%	2%	61%	
Separator						0.0%	0%	0%	28%	
Spot battery cell cost (US\$/kWh)										
LFP	59					0.0%	-1%	-4%	12%	
NMC532	79					0.0%	-2%	-7%	12%	
NMC622	75					0.0%	-2%	-7%	8%	
NMC811	78					0.0%	-1%	-6%	11%	
Spot battery pack cost (US\$/kWh)										
LFP	71					0.0%	-1%	-4%	-4%	
NMC532	110					0.0%	-1%	-5%	9%	
NMC622	108					0.0%	-1%	-5%	5%	
NMC811	111					0.0%	-1%	-4%	8%	

Source: Bloomberg, Baiinfo, Bernstein analysis and estimates (cell and pack cost)

EXHIBIT 2: Key companies price performance and valuation

	Price		Mkt cap	P/E	EV					
	2-Oct	Ccy	\$bn	26E	/EBITDA	%1D	%1W	%1M	%1Y	1Y px
					26E					chart
Cell makers										
LGES	364,000	KRW	62.7	n.a.	15.2x	1.0%	4%	-1%	-9%	
Samsung SDI	525,000	KRW	31.1	76.7x	15.8x	1.5%	1%	-6%	154%	
CATL	291.1	CNY	204.5	14.1x	9.7x	1.5%	-1%	-19%	-28%	
CALB	16.2	HKD	3.7	9.3x	2.2x	-6.7%	-6%	-5%	-55%	
EVE Energy	49.8	CNY	16.1	14.8x	9.4x	2.5%	2%	-5%	-45%	
Gotion	29.5	CNY	8.0	27.0x	6.5x	2.9%	10%	11%	-37%	
Sunwoda	22.0	CNY	6.1	20.3x	6.5x	2.3%	10%	28%	-35%	
Farasis	9.2	CNY	1.7	n.a.	n.a.	1.2%	4%	-1%	-57%	
Lithium Miners/Refiners										
Albemarle	105.0	USD	12.4	9.4x	4.5x	-0.7%	-4%	-23%	20%	
SQM	64.1	USD	17.7	9.0x	4.3x	-1.2%	-4%	-20%	44%	
Pilbara Minerals	3.8	AUD	8.4	20.6x	10.2x	2.9%	-2%	-31%	51%	
Ganfeng Lithium	29.8	HKD	12.3	7.0x	6.7x	-3.4%	-4%	-26%	-38%	
Tianqi Lithium	40.3	CNY	9.7	9.9x	3.5x	-0.7%	-2%	-18%	-15%	
Cathode										
LG Chem	258,500	KRW	13.4	n.a.	2.4x	0.6%	2%	-8%	-7%	
EcoPro BM	114,600	KRW	8.5	580.5x	56.0x	-0.1%	9%	3%	1%	
L&F	112,300	KRW	3.4	n.a.	15.3x	-1.0%	3%	-13%	52%	
Umicore	20.6	EUR	5.7	11.1x	4.9x	-2.7%	-4%	-11%	34%	
Ningbo Ronbay	23.5	CNY	2.5	41.9x	11.2x	1.8%	1%	-8%	-27%	
Beijing Easpring	38.9	CNY	3.2	18.9x	12.1x	2.3%	4%	-3%	-41%	
GEM	6.1	CNY	4.6	16.3x	5.1x	1.2%	0%	-6%	-27%	
Huayou Cobalt	35.0	CNY	9.9	8.6x	3.8x	2.9%	-1%	-11%	-47%	
Dynanonic	39.6	CNY	1.7	13.2x	5.5x	0.8%	-1%	-11%	-14%	
Anode										
Posco	185,000	KRW	12.1	538.4x	49.7x	1.3%	3%	-2%	26%	
Putailai	21.5	CNY	6.9	14.1x	8.8x	0.8%	0%	-7%	-31%	
Shanshan	11.5	CNY	3.9	n.a.	n.a.	-0.6%	0%	-3%	-28%	
Separator										
Asahi Kasei	1,622	JPY	14.0	15.0x	5.5x	-1.7%	-1%	-4%	40%	
SK IE	17,160	KRW	1.0	n.a.	-21.1x	5.7%	0%	1%	-36%	
Toray Industries	1,221	JPY	11.6	22.7x	7.1x	-0.9%	-3%	-7%	30%	
W-Scope	141	JPY	0.1	n.a.	n.a.	-2.1%	-5%	-12%	-36%	
Yunnan Energy	45.5	CNY	6.7	20.1x	7.7x	-0.2%	-2%	-13%	-3%	
Shenzhen Senior	15.7	CNY	3.3	32.5x	10.2x	-1.7%	-10%	10%	8%	
Sinoma Science	54.2	CNY	13.5	30.0x	12.6x	-1.2%	-5%	6%	59%	
Electrolyte										
Enchem	22,700	KRW	0.4	n.a.	n.a.	5.3%	12%	12%	-67%	
Tinci Minerals	31.2	CNY	9.5	10.1x	7.1x	0.4%	-2%	-12%	-18%	
Capchem	66.5	CNY	7.5	22.3x	16.6x	0.9%	-6%	3%	25%	
Jiangsu Guotai	7.6	CNY	1.8	9.6x	n.a.	1.5%	1%	-3%	-17%	
Shanshan	11.5	CNY	3.9	n.a.	n.a.	-0.6%	0%	-3%	-28%	
Equipment Maker										
Hirano	1,781	JPY	0.2	16.8x	n.a.	-2.1%	-3%	-5%	12%	
CKD	5,730	JPY	2.5	30.8x	16.0x	1.6%	9%	8%	91%	
Wuxi Lead	34.9	CNY	8.5	23.8x	18.9x	3.5%	9%	6%	-44%	
Shenzhen Yinghe	21.4	CNY	2.1	20.7x	12.5x	1.2%	2%	0%	-38%	
Zhejiang HangKe	20.4	CNY	1.8	23.1x	18.9x	0.3%	-2%	-5%	-44%	
Guangdong Lyric	42.2	CNY	1.1	45.6x	18.6x	1.6%	12%	24%	-42%	
Shenzhen United	20.1	CNY	1.0	20.4x	12.5x	-1.0%	-5%	-4%	-24%	

Source: Bloomberg (all estimates), Bernstein analysis
GLOBAL ENERGY STORAGE

Company Models

[CATL](#) | [LG Energy Solution](#) | [Samsung SDI](#)

[LG Chem](#) | [EcoPro BM](#) | [Posco Future M](#)

[Tianqi Lithium](#)

[Sungrow](#)

Industry Models

[Model: ESS tracker](#)

[Model: Monthly EV Tracker](#)

[Model: Battery Gigafactory Database](#)

[Industry model: Global EV TAM](#)

[Industry model: Global ESS TAM](#)

[Industry model: Battery TAM](#)

[Industry model: Battery supply and demand](#)

Blackbook

[Battery Utopia](#)

[Battery Tech Roadmap 2023: Faster, Higher, Stronger](#)

[Turn Up the Voltage... Investing Across the Battery Value Chain](#)

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[Global Energy Storage: Initiating Coverage. Lithium Positive \(Tianqi\) and Negative Cathodes \(EcoPro BM, Posco Future M\)](#)

[Sungrow: Solar Eclipse - Initiating at Market-Perform \(PT RMB65\)](#)

Recent Research

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[CATL: A winner in the age of energy disruption](#)

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[CATL: Initial perspectives on HK IPO Plans](#)

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[Global Energy Storage: Bigger and better batteries enables the rise of the 700km electric vehicle](#)

[Global Energy Storage: 2025 Lithium Outlook. Rising from rock bottom](#)

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BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	30 Sep 2026		TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
			Closing Price	Price Target		Cur	2025A	2026E	2027E	2025A	2026E	2027E
300750.CH (CATL)	O	CNY	291.11	800.00	(47.6)%	CNY	16.14	21.95	28.77	18.0	13.3	10.1
3750.HK (CATL)	O	HKD	484.20	770.00	(40.2)%	CNY	16.14	21.95	28.77	25.6	18.9	14.4
051910.KS (LG Chem)	M	KRW	257,000	312,000	(28.2)%	KRW (13,258.70)	2,693.22	28,743	(19.4)	95.4	8.9	
373220.KS (LGES)	M	KRW	360,500	351,000	(30.5)%	KRW (4,580.94)	(2,364.33)	10,225	(78.7)	(152.5)	35.3	
006400.KS (SDI)	M	KRW	517,000	520,000	129.5%	KRW (9,933.80)	1,751.66	18,577	(52.0)	295.1	27.8	
247540.KS (EcoPro BM)	U	KRW	114,700	140,000	(19.5)%	KRW	403.00	854.00	1,963.00	284.6	134.3	58.4
300274.CH (Sungrow)	O	RMB	82.49	151.00	(69.9)%	RMB	6.55	6.93	7.95	12.6	11.9	10.4
002466.CH (Tianqi Lithium)	O	CNY	40.25	80.00	(36.2)%	CNY	0.28	3.90	7.75	142.6	10.3	5.2
9696.HK (Tianqi Lithium)	O	HKD	28.68	65.00	(64.3)%	CNY	0.28	3.90	7.75	86.9	6.3	3.2
ASIA			1,948.97									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

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